

CaféPulse

Enterprise Real-Time Inventory & Multi-Tenant Supply Chain Operating System for Specialty Coffee Chains

Document Reference	SRS-CAFEPULSE-2026-V1.0
Standard Compliance	IEEE Std 830-1998 / ISO/IEC/IEEE 29148:2018
Software Version	Version 1.0.0 (Production Hardened)
Target Cloud Engine	Google Cloud Firestore (ID: caf247-73960)
Framework & Runtime	Next.js 15.1 (App Router) / React 19 / Node.js
Primary Currency	Indian Rupee (INR ₹ / en-IN)
Release Date	August 2026
Security & Access	Multi-Role RBAC & Isolated Data Isolation

1. EXECUTIVE SUMMARY & DOCUMENT CONTROL

CaféPulse is a real-time, multi-tenant inventory control and supply chain orchestration platform engineered specifically for specialty coffee roasteries, artisan espresso bars, and multi-location café franchises. The system bridges physical café floor operations (daily usage steppers, instant restocks, wastage logs) with public wholesale B2B supplier bidding, automated PO creation, price agreement contract mapping, and comprehensive audit accounting in Indian Rupees (₹).

1.1 Document Overview & Standards

This document serves as the formal **Software Requirements Specification (SRS)** compliant with **IEEE Std 830-1998** and **ISO/IEC/IEEE 29148:2018**. It delineates all functional capabilities, non-functional performance benchmarks, data models, role-based security policies, and user acceptance criteria implemented across the system.

1.2 Glossary & Acronyms

TERM / ACRONYM	DEFINITION & OPERATIONAL CONTEXT
COGS	Cost of Goods Sold — The direct valuation of ingredients utilized in coffee preparation and kitchen production.
PAR Level	Periodic Automatic Replenishment — The optimal target inventory ceiling required to satisfy expected demand.
Reorder Threshold	The minimum safety buffer that automatically triggers restock alerts and PO generation recommendations.
SKU	Stock Keeping Unit — Unique machine-readable identifier generated per category and item.
PO	Purchase Order — Legally binding B2B procurement order issued from a café branch to an approved supplier.
MOQ	Minimum Order Quantity — Supplier-enforced minimum batch size for wholesale replenishment.
RBAC	Role-Based Access Control — Permissions framework enforcing access tiers across Store Admin, Shift Barista, Auditor, and Vendor.
Goods Receipt (GRN)	The physical verification and barcode inspection step that checks received deliveries into live stock.

2. SYSTEM ARCHITECTURE & DATABASE DESIGN

CaféPulse operates on a serverless, event-driven architecture combining Next.js 15 App Router clients with **Google Cloud Firestore** for ACID-compliant real-time multi-document data synchronization.

2.1 Cloud Firestore Multi-Tenant Schema

COLLECTION NAME	DOCUMENT KEY STRATEGY	CORE ATTRIBUTES & RELATIONSHIPS
users	uid (Firebase Auth UID)	displayName, cafeName, branchName, branchCode, city, role, monthlyVolumeEstimate, status
inventory_items	\${userId}_\${itemId}	name, sku, categoryId, quantity, unit, packageWeight, unitPrice, reorderLevel, parLevel, supplier, vendorMappings[]
categories	\${userId}_\${catId}	name, description, color, icon, userId
purchase_orders	po-\${timestamp}-\${rand}	poNumber, userId, destinationCafe, supplierName, status (PENDING_DELIVERY DELIVERED), items[], totalEstimatedCost
vendors	\${userId}_\${vendorId}	name, contactPerson, email, phone, city, gstin, leadTimeDays, paymentTerms
supplier_applications	app-\${timestamp}-\${rand}	companyName, targetCafeId, targetCafePreference, quotedItems[], status (PENDING APPROVED REJECTED)
staff_members	staff-\${timestamp}-\${rand}	name, email, phone, role, branch, pin, status (ACTIVE INACTIVE)
activity_logs	log-\${timestamp}-\${rand}	action, targetId, itemName, detail, user, timestamp

2.2 Vendor Price Book Mapping Schema

```
// Sub-schema: vendorMappings embedded inside inventory_items document { "mappingId": "vm-1723932840-a1b2",  
"vendorName": "Mercara Coffee Roasters", "vendorItemName": "Chikmagalur Estate Arabica (AAA Roaster Cut)",  
"vendorSku": "MCR-AAA-01", "unitPrice": 1450.00, "isPreferred": true, "leadTimeDays": 2, "notes": "Direct farm  
agreement, Net 15 terms" }
```

2.3 Offline Fallback & Dual-Bus Synchronization Engine

To guarantee uninterrupted café point-of-sale and floor operations during internet connectivity dropouts, the system utilizes a **Dual-Bus Synchronization Engine**:

- **Client Event Bus:** Emits local in-memory state updates instantly (0ms latency) to guarantee seamless, zero-lag UI feedback.
- **Local Storage Cache:** Persists state snapshots scoped per `userId` under isolated keys (`cafepulse_items_${uid}`).
- **Firestore Real-Time Listener:** Subscribes via `onSnapshot` using `memoryLocalCache()` to keep all multi-tab clients synchronized while completely eliminating lease-lock warnings.

3. DETAILED FUNCTIONAL REQUIREMENTS (FR)

3.1 Real-Time Stock Management & Inventory Catalog (FR-INV)

REQ ID	FEATURE NAME	FUNCTIONAL DESCRIPTION & VALIDATION RULES
FR-INV-01	Live Stepper Adjustments	Staff can increase (Restock) or decrease (Usage) stock directly from tables with single clicks. Atomic increments are applied to Firestore.
FR-INV-02	Automatic SKU Generation	System automatically creates structured SKUs based on Category prefix, item name syllables, and pseudo-random numbers (e.g. COF-CHIK-428).
FR-INV-03	Dynamic Stock Status	Items are dynamically classified as In Stock (Green), Low Stock (Amber, when $\leq$ Reorder Level), or Out of Stock (Red, when $= 0$).
FR-INV-04	CSV Export Engine	Managers can export full or filtered catalog views to standard RFC 4180 CSV with formatted INR values and stock health tags.

3.2 Multi-Tenant Public Sourcing & B2B Vendor Proposals (FR-B2B)

REQ ID	FEATURE NAME	FUNCTIONAL DESCRIPTION & VALIDATION RULES
FR-B2B-01	Real-Time Café Discovery	Public supplier portal (<code>/supplier/register</code>) discovers all active registered cafés from Firestore <code>users</code> collection in real-time.
FR-B2B-02	Store-Specific Sourcing Demands	Selecting a café dynamically loads that café's actual inventory items for wholesale rate card quoting.
FR-B2B-03	Custom Product Proposals	Suppliers can pitch custom, unlisted ingredients via <code>+ Propose Custom Product</code> with custom MOQ, SKU, unit, and wholesale rates.
FR-B2B-04	Targeted Proposal Inbox	Proposals route directly to the designated café manager's inbox on <code>/dashboard/vendors</code> based on <code>targetCafeId</code> .
FR-B2B-05	1-Click Approval & Auto-Catalog	Approving a proposal registers the vendor, updates price books, and auto-creates any proposed custom items into the store's inventory catalog.

3.3 Automated Shortage Radar & Purchase Orders (FR-ORD)

REQ ID	FEATURE NAME	FUNCTIONAL DESCRIPTION & VALIDATION RULES
FR-ORD-01	Automated Shortage Radar	Identifies items below safety thresholds and calculates optimal restock quantities: $\text{Suggested} = \text{PAR} - \text{CurrentQty}$.
FR-ORD-02	1-Click Bulk PO Generation	Aggregates low-stock items grouped by supplier and auto-populates pre-filled Purchase Orders with contracted price agreements.
FR-ORD-03	Supplier Fulfillment & Dispatch	Suppliers log in to <code>/supplier/portal</code> , inspect incoming PO line items, and mark orders as Dispatched with tracking numbers.
FR-ORD-04	Goods Receipt (GRN) Restock	When physical goods arrive at the café, managers click Receive Goods to automatically increment catalog inventory counts.

4. NON-FUNCTIONAL & SECURITY REQUIREMENTS

4.1 Performance & Latency Benchmarks (NFR-PERF)

- **Client UI Response Time:** Immediate local optimistic UI update within $\leq 16\text{ms}$ (60 FPS animation budget).
- **Cloud Firestore Latency:** Remote document synchronization across active clients within $\leq 350\text{ms}$ under standard 4G/Broadband.
- **Cold Start & Static Page Load:** First Contentful Paint (FCP) $\leq 0.8\text{s}$ across all static routes.

4.2 Role-Based Access Control (RBAC) Matrix (NFR-SEC)

USER ROLE	INVENTORY CATALOG	ADJUST STOCK	PURCHASE ORDERS	FINANCIALS / COGS	STORE DANGER ZONE
 Store Admin / GM	FULL CRUD	FULL ACCESS	ISSUE & RECEIVE	FULL ACCESS	FULL ACCESS
 Head Barista	EDIT ITEMS	RESTOCK & WASTE	DRAFT & INSPECT	RESTRICTED	NO ACCESS
 Shift Barista	READ-ONLY	QUICK USAGE (-1)	NO ACCESS	HIDDEN	NO ACCESS
 Inventory Auditor	READ-ONLY	NO ACCESS	AUDIT TRAILS	FULL VALUATION	NO ACCESS

4.3 Data Retention & Account Deletion (NFR-PRG)

Zero Data Retention Policy on Deletion: When an administrator triggers store deletion via `/dashboard/team`, all associated documents in `users`, `inventory_items`, `categories`, `purchase_orders`, `supplier_applications`, `vendors`, `staff_members`, and `activity_logs` are permanently purged via atomic multi-document deletes. The store is immediately unlisted from the public directory.

5. VERIFICATION & ACCEPTANCE TEST MATRIX

TEST ID	TEST SCENARIO	TARGET BEHAVIOR & ACCEPTANCE STATUS	RESULT
TC-AUTH-01	New Café Registration	New account on <code>/login</code> saves profile to <code>users</code> and appears dynamically on <code>/supplier/register</code> .	PASSED
TC-B2B-01	Supplier Bidding on Café	Selecting café loads that café's catalog; bids route to that specific manager's vendor inbox.	PASSED
TC-B2B-02	Custom Item Auto-Catalog	Approved custom product creates document in <code>inventory_items</code> with mapped price agreement.	PASSED
TC-ORD-01	PO Dispatch & GRN	Supplier dispatches PO with tracking; Café receiving goods increments item stock count in Firestore.	PASSED
TC-PRG-01	Store Account Deletion	Danger zone delete purges all Firestore collections for that UID and removes store from public list.	PASSED
TC-BLD-01	Production Build	<code>next build</code> executes with 0 TypeScript/JSX errors across all 16 static routes.	PASSED

Sign-Off & Document Approval

Software Architecture & Production Release Confirmed • Version 1.0.0 • August 2026